

PRACTICAL 8-

AIM- A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

CODE-

```
GNU nano 8.7                                     page.c
#include<stdio.h>

int main()
{
    int pages[12] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frames[3];
    int i,j,found;
    int hits = 0, faults = 0;

    for(i=0;i<3;i++)
        frames[i] = -1;

    for(i=0;i<12;i++)
    {
        found = 0;

        for(j=0;j<3;j++)
        {
            if(frames[j] == pages[i])
            {
                hits++;
                found = 1;
                break;
            }
        }

        if(found == 0)
        {
            frames[i%3] = pages[i];
            faults++;
        }
    }

    printf("Page Hits = %d\n",hits);
    printf("Page Faults = %d\n",faults);

    printf("Hit Percentage = %.2f\n",(hits/12.0)*100);
    printf("Fault Percentage = %.2f\n",(faults/12.0)*100);

    return 0;
}
```

OUTPUT-

```
HP@DESKTOP-B4BD73N MINGW64 ~  
$ nano page.c  
  
HP@DESKTOP-B4BD73N MINGW64 ~  
$ gcc page.c  
  
HP@DESKTOP-B4BD73N MINGW64 ~  
$ ./a.out  
-bash: ./a.out: No such file or directory  
  
HP@DESKTOP-B4BD73N MINGW64 ~  
$ gcc page.c -o page  
  
HP@DESKTOP-B4BD73N MINGW64 ~  
$ ./page  
Page Hits = 2  
Page Faults = 10  
Hit Percentage = 16.67  
Fault Percentage = 83.33
```